Ashish Makani

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Research interests: medical and clinical AI; agentic AI systems and human–AI teams for biomedical discovery; evaluation and auditing of large language models; digital health for improving access to and delivery of healthcare.

Publications

NeurIPS 2026 Workshop Papers *3 papers accepted, NeurIPS 2026 (Sydney)*

[1] **A Voice-Enabled Palliative Care Assistant with Clinician Review and Preliminary Expert Evaluation.**
The Third Workshop on GenAI for Health: Agentic Systems, Clinical Trust, and Future Potential (GenAI4Health), NeurIPS 2026. [NeurIPS workshop page]

[2] **Auditing Longitudinal LLM Evaluations with First-Failure Survival: A Synthetic Study.**
The Third Workshop on GenAI for Health: Agentic Systems, Clinical Trust, and Future Potential (GenAI4Health), NeurIPS 2026. [NeurIPS workshop page]

[3] **Patients Are Waiting: From Swarms to Synergy. Designing Complementary Collaborative Human–AI Agent Teams for Biomedical Discovery & Invention.**
Agentic AI for Biological Discovery Workshop (Agentic LS), NeurIPS 2026. [NeurIPS workshop page]

Preprints

[4] **A. Makani**, Anjali Agrawal, Anurag Agrawal. **Onco-Seg: Adapting Promptable Concept Segmentation for Multi-Modal Medical Imaging.**
medRxiv, 2026. doi:10.64898/2026.01.11.26343874

[5] **A. Makani. Onco-Shikshak: An AI-Native Adaptive Learning Ecosystem for Medical Oncology Education.**
medRxiv, 2026. doi:10.64898/2026.02.23.26346944

Grants & Funding

Lead Applicant, EVAH grant (US$1 million), J-PAL; funded by Wellcome Trust, UK
Status: selected for funding; formal award in process

- Large-scale clinical evaluation and nationwide randomised controlled trial (RCT) of **Palli Sahayak**, a voice-enabled palliative care assistant; led by Ashish Makani at Ashoka University.
- *Clinical consortium (4 sites):* CMC Vellore; KMC Manipal (MAHE); Coimbatore Cancer Foundation (CCF); Cachar Cancer Hospital & Research Centre (CCHRC).
- *Regulatory partners:* WHO SEARO; ICMR-NIRDHDS.

Lead Architect & Developer, Grand Challenges India (GCI) Phase 1 grant (US$100,000)
Awarded by the Gates Foundation and DBT–BIRAC

- **Palli Sahayak**: open-source, voice-first palliative care assistant answering questions by phone in Indian languages from curated care guidance; phase-one expert review by palliative care physicians from Pallium India and Max Healthcare.

Selected, Grand Challenges India (GCI) Phase 2: scale-up and adaptation of Palli Sahayak

- **Vaani Sakhi**: clinically grounded voice-AI agent for women's health, starting with postmenopausal care (Hindi and English), built on the Palli Sahayak engine.

Selected Research Projects & Recognition

AI-Powered Virtual Tumor Board

- A panel of specialist AI agents that reviews a cancer case the way a multidisciplinary tumour board would, to bring expert, guideline-based treatment planning to patients far from major cancer centres.
- *Press:* profiled with the Virtual Tumor Board by the *Times of India* ("Times Techies: AI for Social Impact"), January 2026.

Onco-Seg

- Adapts promptable concept segmentation (SAM3) for multi-modal medical image segmentation, delineating tumours and organs across CT, MRI, ultrasound and more (preprint [4]).
- Highlighted by Meta's head of AI at the AI Impact Summit, New Delhi, February 2026.

Onco-Shikshak

- Retrieval-augmented, AI-native adaptive learning platform for medical oncology education: six specialist AI agents, guideline-grounded RAG, spaced repetition and adaptive difficulty (preprint [5]).

Med Abhyaas

- Practice copilot for medical education: virtual patient encounters, including in Indic languages, for developing clinical reasoning, patient communication and empathy (research in progress).

Essays & Writing

- **Curing All Disease, Ahead of Schedule: Permission to Dream Bigger with Complementary AI–Human Teams**, Nov 2026. What the AI hub-and-spoke model means for pharma, CROs and frontier labs, and why humans must stay at the centre of discovery.
- **Some Thoughts on Designing Optimal Swarm Architectures: The Notion of the Collective Good in Swarms**, Nov 2026. How to get a swarm of AI agents to act for the collective good without coordination overhead overwhelming the main task.
- **AI × Science** (living page): an ongoing reading list on AI agents, scientific discovery, mathematics, biology and medicine.

More essays: inventcures.github.io/essays

Research Experience

KCDH-A (Koita Centre for Digital Health at Ashoka), Ashoka Univ. Sonipat, India
Program Manager, Research Jan 2026 – present

Centre for Bioinformatics & Computational Biology, Ashoka Univ. Sonipat, India
Project Scientist May 2023 – Dec 2025

- Helped lead the Centre for Bioinformatics & Computational Biology under the faculty PIs.
- Performed research on medical AI; started a journal club on medical AI.
- Organised training events and workshops as part of the Centre's programming.

Liberin Technologies Noida, India
Senior Consultant (Python & Machine Learning) Apr 2019 – Nov 2019

- Surveyed and reviewed the state of the art in machine learning for health, with a focus on machine learning for radiology.

CARING Research, Mahajan Imaging New Delhi, India
Senior Development Engineer Oct 2018 – Apr 2019

- Led deployment of CARING's AI solution RAID at two large hospitals in Mumbai.

- Led the lung segmentation project using open-source 3D Slicer and the Chest Imaging Platform (CIP).
- Managed Linux servers and containerised (Docker) machine learning models; wrote Python tooling for manipulation and analysis of radiology research data.

National Centre for Biological Sciences (NCBS) Bengaluru, India
Junior Research Fellow Jan 2018 – Apr 2018

- Data analysis for research in Python (NumPy, pandas, scikit-learn).

Gram Vaani, IIT Delhi New Delhi, India
Project Associate Aug 2012 – Mar 2013

- Implemented a monitoring framework for the production environment, with automated tests and failure notifications.

SETLabs (R&D), Infosys Technologies Bengaluru, India
Research Engineer Jan 2004 – Dec 2006

- Research in computer and wireless networks.
- Initiated and led the Wireless Sensor Network (WSN) project, an integrated industrial temperature and pressure monitoring application, across the full lifecycle from market survey and vendor selection through testbed experimentation, prototyping and integration testing.
- Played a leadership role in a cross-organisational Infosys–Intel team that developed a secure architecture for deploying Wi-Fi across the Infosys Bengaluru campus.

Industry Experience

Xebia IT Architects Gurugram, India
Lead Consultant Nov 2019 – Jun 2022

- *Levi Strauss (Apr 2020 – Jun 2022):* validated AWS components (Lambda, data pipelines, state machines, batch jobs); wrote and validated SQL for evolving business logic; built test automation in Java (Selenium, AWS SDK).
- *Chegg (Nov 2019 – Apr 2020):* built UI and back-end test automation in Python (Selenium, Robot Framework).

Intel Technology India (via HCL Technologies) Bengaluru, India
Software Engineer Jul 2013 – Aug 2015

- Tested Intel's tablet SoC audio device driver in Python; led the team's transition to a new automation framework and trained other tablet teams on it.
- Piloted an internal WHCK automation tool that sharply reduced Windows logo certification effort.
- Built a TeamCity continuous-integration pipeline automating the audio driver validation workflow.

Telecom Technology Solutions Pleasanton, CA, USA
RF Engineer Aug 2010 – Jan 2011

- RF design and optimisation for cellular operators (T-Mobile, AT&T); drive tests, coverage and interference analysis.

NextWave Wireless San Diego, CA, USA
Intern, Systems Engineering Jun 2008 – Aug 2008

- Implemented in C++ a new configuration file format and parser for the WiMAX system simulator.

Education

Purdue University West Lafayette, IN, USA
M.S., Electrical & Computer Engineering Dec 2009

Jamia Millia Islamia (JMI), Faculty of Engineering & Technology New Delhi, India
B.Tech., Electronics & Communication Engineering Jun 2003

- Merit scholarship with full tuition waiver for securing 2nd rank in the university.

Dept. of EECS, University of California, Berkeley Berkeley, CA, USA
Course: Cancer Genomics & Computing (experimental course) Aug 2011 – Nov 2011

- Co-taught by David Patterson and UCSF pathology resident Taylor Sittler; fundamentals of genetics, the genetic causes of cancer, and genomics.

Selected graduate coursework: Probability & Random Variables; Linear Algebra; Statistical Methods for Engineering; Operating Systems; Computer Network Systems; System-on-Chip; Embedded Systems; Digital Logic Synthesis.

Teaching

Purdue University, West Lafayette, IN, USA

- *Teaching Instructor*, CS 314 (Numerical Methods) Jun 2010 – Jul 2010
- *Teaching Assistant*, CS 177 (Intro. to Computer Science; HTML, JavaScript) Jan 2008 – Jun 2008
- *Teaching Assistant*, ENGR 126 (MATLAB programming labs, 3 sections) Aug 2007 – Dec 2007

Recitations, lab sessions, office hours, and creating and grading labs, projects and exams.

Earlier Graduate & Engineering Projects

- **Hardware–software co-design of a JPEG decoder on FPGA** (Altera DE2; C++, Verilog).
- **Wireless sensor network nodes for robot navigation**: compass and IR sensing with Crossbow IRIS nodes to navigate an iRobot Create (nesC, C).
- **A TCP-like transport protocol** improving end-to-end transfer speed with reliability and in-order delivery (Java).
- **Speech recognition on mobile devices via cyber-foraging** (Infosys SETLabs pilot): offloading compute-intensive recognition to a nearby surrogate machine (Java).

Technical Skills

Languages: Python, C, C++, Java, SQL (PL/SQL), HTML/JavaScript, Verilog/VHDL, MATLAB
ML & data: NumPy, pandas, scikit-learn; 3D Slicer, Chest Imaging Platform
Systems & tools: GNU/Linux, Docker, AWS, TeamCity (CI), Selenium, Robot Framework

Service & Outreach

- Volunteer TA, RailsBridge (programming workshops for women in the San Francisco Bay Area).
- Volunteer tutor, Dream Skool Foundation, Bengaluru: math and science for grade 11–12 students from low-income households preparing for engineering entrance exams.
- Volunteer, Office of International Students & Scholars, Purdue University (Fall 2007 orientation).
- Volunteer, Infosys Foundation (notebook drive for children in villages near Bengaluru).

Last updated: November 2026